



Genome Alignment and Variant calling

Shahiid Kiyaga

Learning Objectives

- Understand the principles of variant calling and the importance of validation.
- Learn about key bioinformatics tools and algorithms used for variant calling.
- Gain hands-on experience in performing variant calling using validated workflows.
- Interpret and assess the quality of called variants.

Introduction to genetic variants

- There are approximately 3 billion base pairs in the human genome.
- Humans share 99.5% of DNA with other humans.
- A **variant** is a difference between similar genomes.
- In most cases – this means a difference between DNA sequences compared to a **reference genome**.
- In this context - a variant is described by its location (genomic coordinates) and genetic change.
 - e.g. chr2 9834 A→G

Introduction to genetic variants

- There is high degree of similarity but the human genome is large ~3 billion nucleotides.
- This results in approximately 4-5 million variants between any individual and the reference genome.
- These, seemingly small number of variations likely explains a significant proportion of phenotypic diversity among humans.



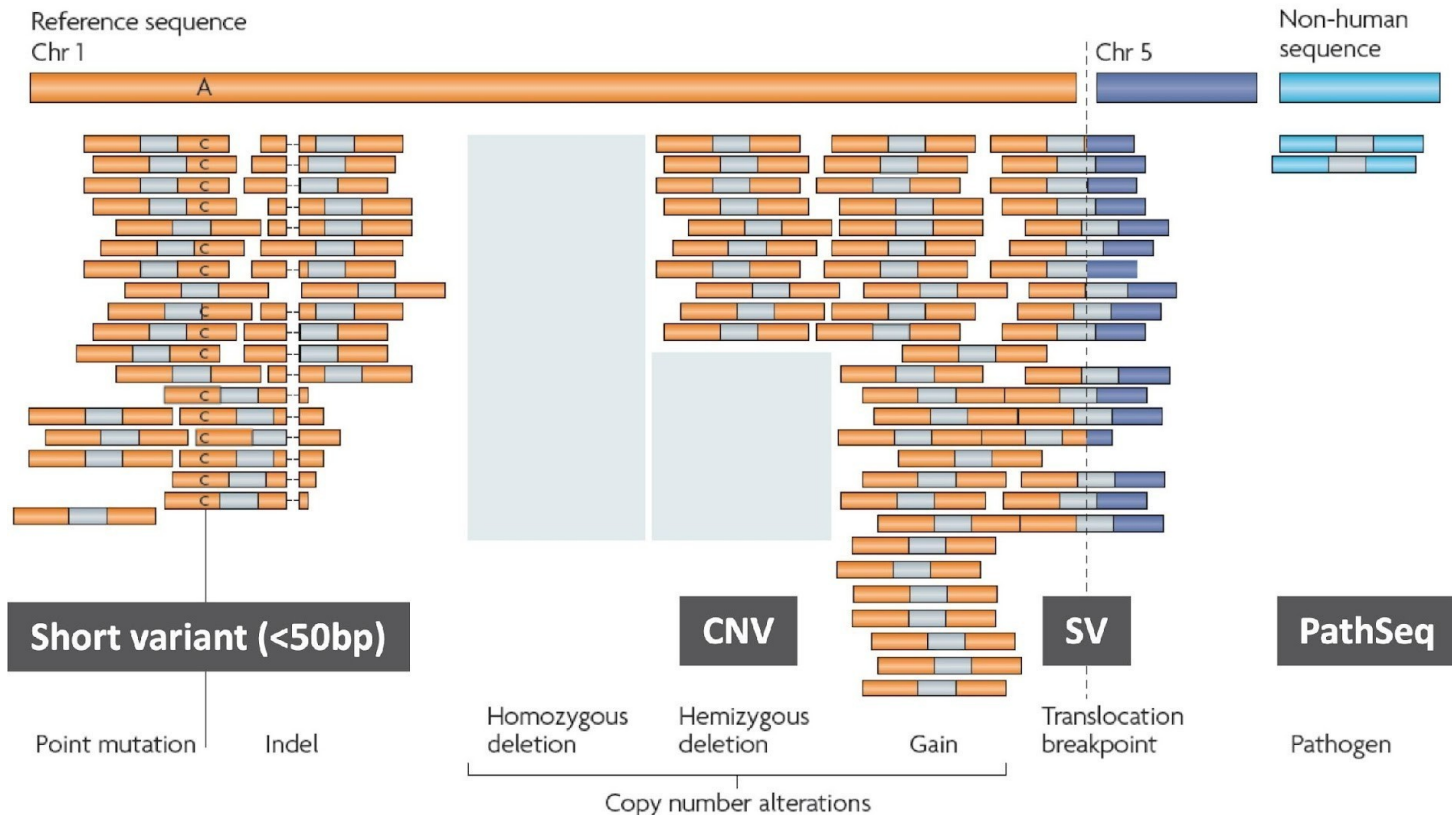
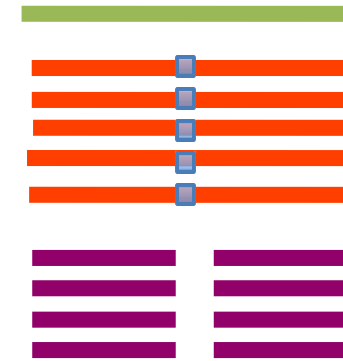
Types of genetic variants

- Single Nucleotide Variants (**SNVs**): Single base substitutions.
- Insertions and Deletions (**INDELs**): Small additions or removals of bases.
- Copy Number Variations (**CNVs**): Duplications or deletions of larger segments.
- Structural Variants (**SVs**): Large chromosomal rearrangements, such as inversions or translocations.

Reference

SNV

Small indel



Types of genetic variants

Single Nucleotide Polymorphism (SNP)

- A single base change (A, T, C, or G) in the DNA sequence that occurs at a specific position in the genome and is common in the population/study samples (typically found in >1% of individuals).
- This mutation at one nucleotide position in the pathogen's genome can:
 - Differentiate strains
 - Affect virulence, drug resistance, or transmission
 - Be used to track outbreaks or evolutionary relationships

Challenges in Variant Calling

- ☐ **Sequencing Errors:** Can introduce false positives.
- ☐ **Read Alignment Issues:** Incorrect mapping affects variant detection.
- ☐ **Low Variant Frequencies:** Rare variants are harder to detect.
- ☐ **Complex Genomic Regions:** Repetitive sequences complicate variant calling.
- ☐ **Data Quality:** Poor-quality sequencing data leads to unreliable results.

Importance of Using Validated Tools

- ▶ Ensures **high accuracy** and reduces false positives.
- ▶ Standardized workflows improve **reproducibility**.
- ▶ Optimized algorithms handle **sequencing errors and biases**.
- ▶ Well-documented and tested tools facilitate **clinical and research applications**.

Variant Calling Workflow

1. Pre-processing

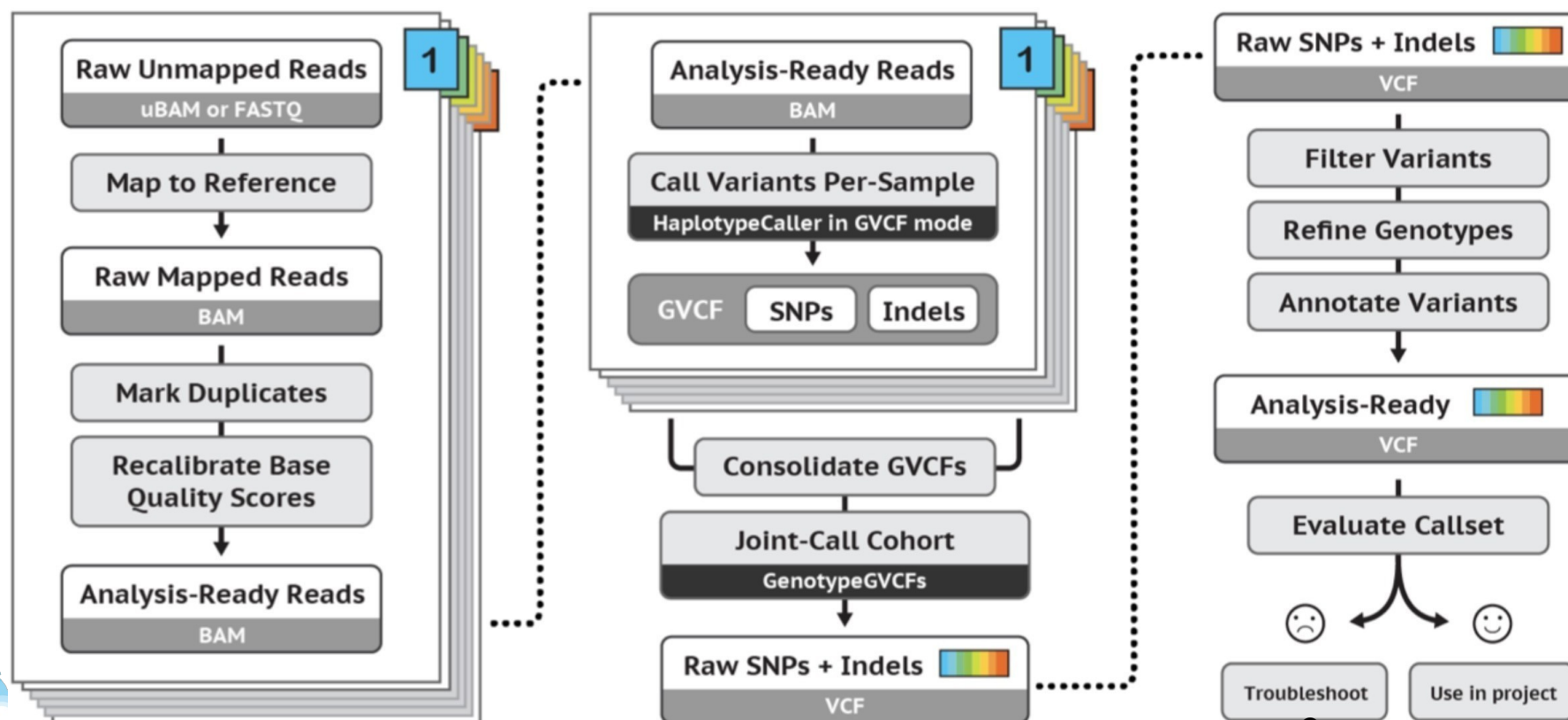
- Prepare analysis ready bam files

2. Variant discovery

- Variant discovery

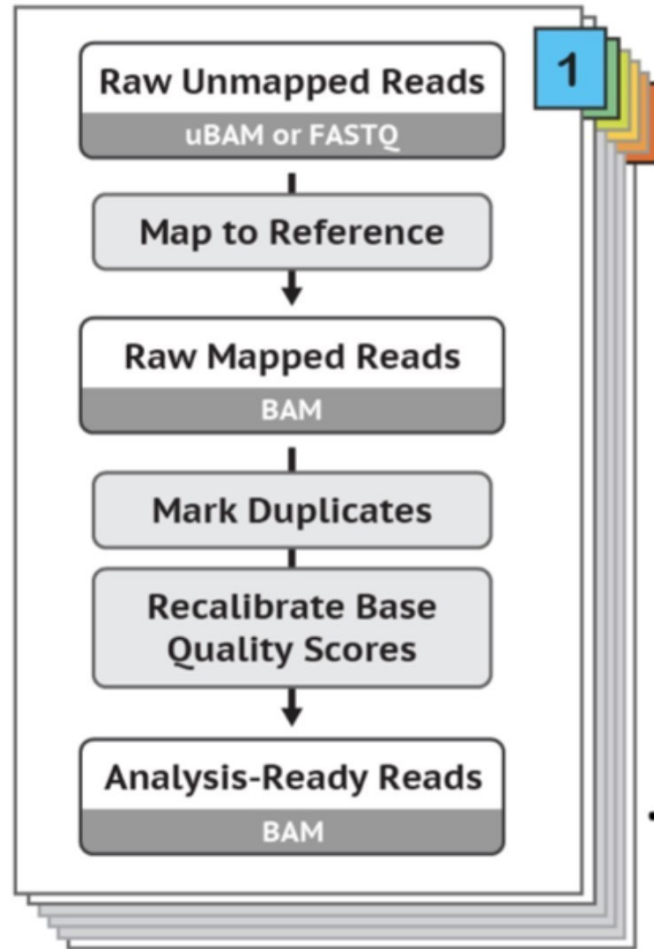
3. Refinement

- Use metadata from variant calling to improve genotype accuracy



Variant Calling Workflow: Pre-processing

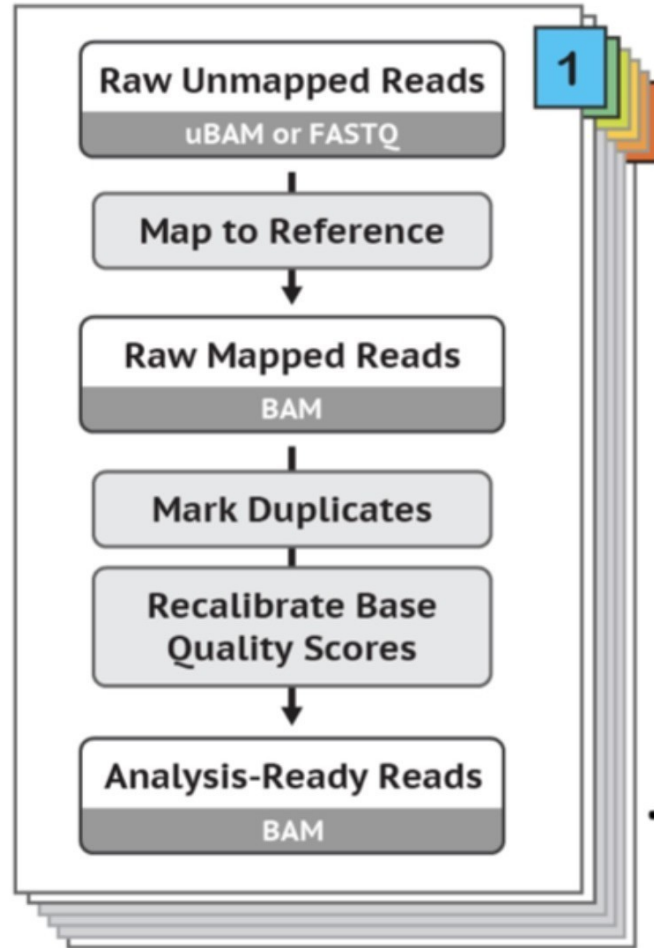
Pre-processing



- A sequencing experiment results in a large volume of sequencing reads.
- Reads are not mapped to a reference and can contain errors and technical artifacts.
 - e.g. a molecule sequenced multiple times will result in duplicate reads

Variant Calling Workflow: Pre-processing

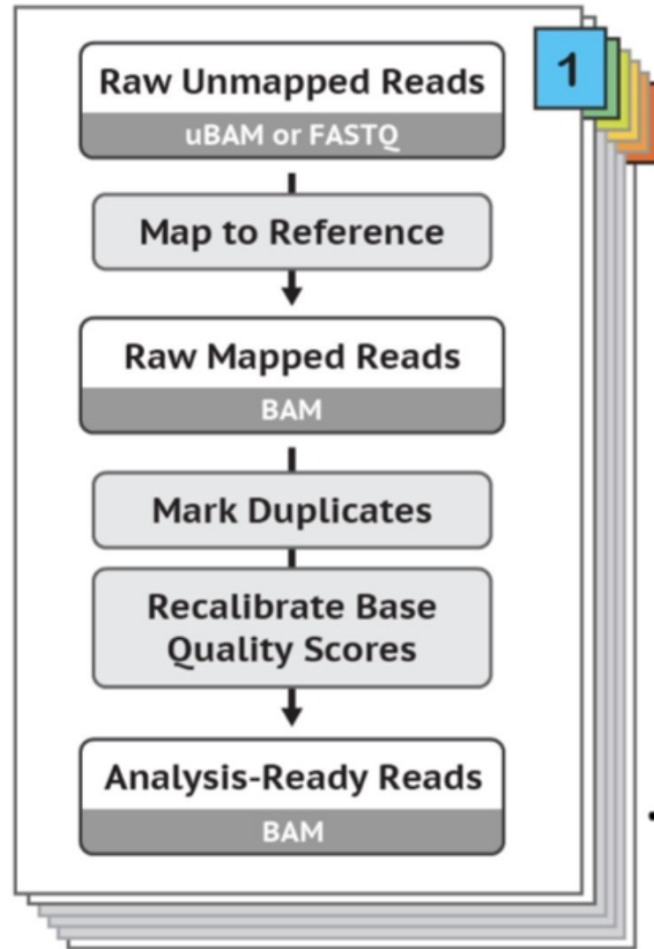
Pre-processing



- We need to filter and prepare the reads and the alignment data – ready for variant calling
- Ensures high-quality input data before variant calling.
- Common steps: Quality control, trimming, and alignment.
- Tools: FastQC, Trimmomatic, BWA, Bowtie2.

Variant Calling Workflow: Pre-processing

Pre-processing

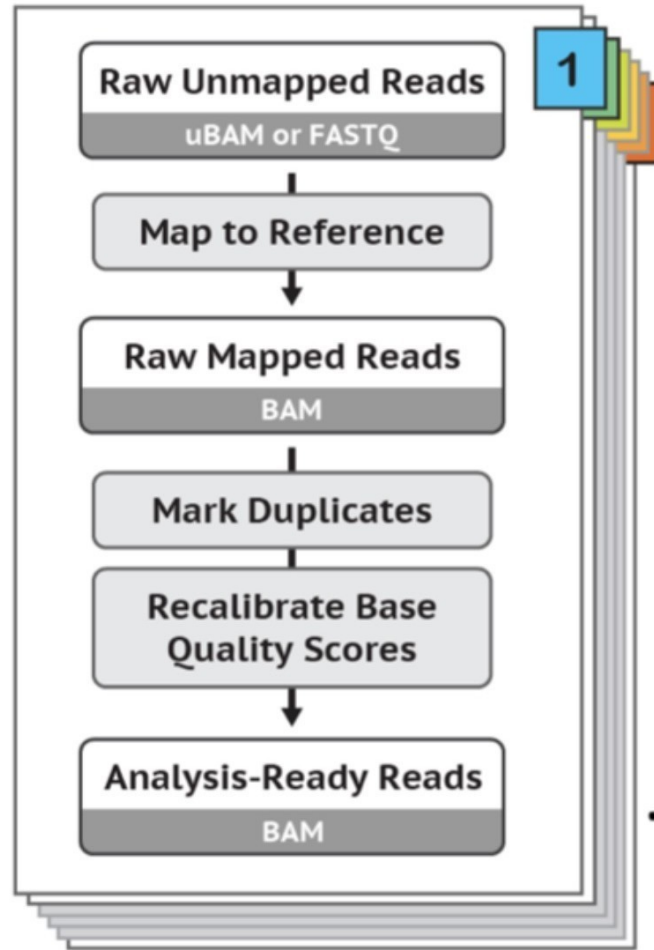


Read Trimming

- Removes low-quality bases and adapter sequences
- Improves the accuracy of read alignment and variant calling.
- Command-line tool with flexible trimming parameters
- Tools: Trimmomatic, Cutadapt, Fastp, etc – NGS
- Porechop, qcat, Guppy Barcoder, NanoFilt, Filtlong

Variant Calling Workflow: Pre-processing

Pre-processing

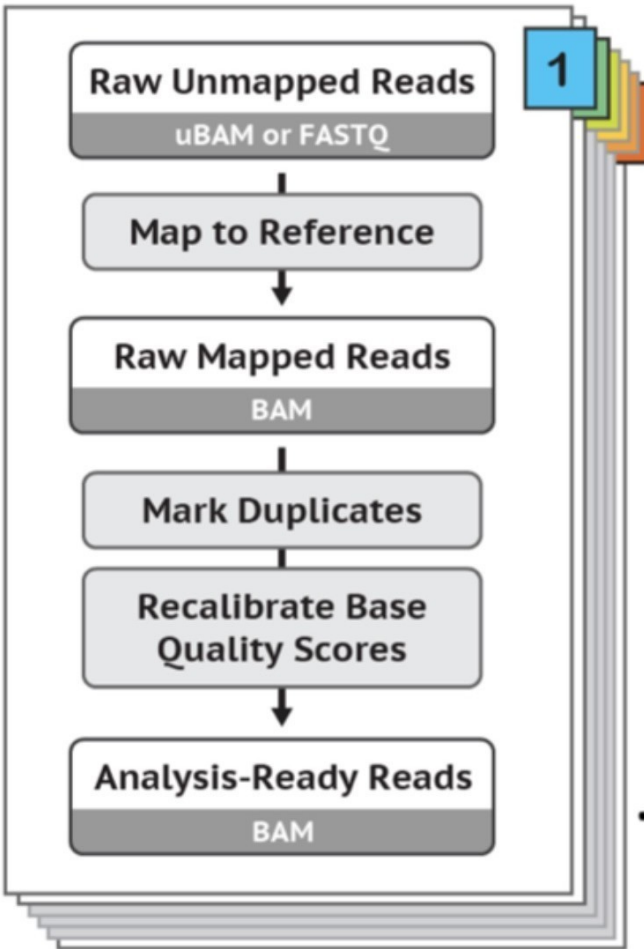


Map to reference

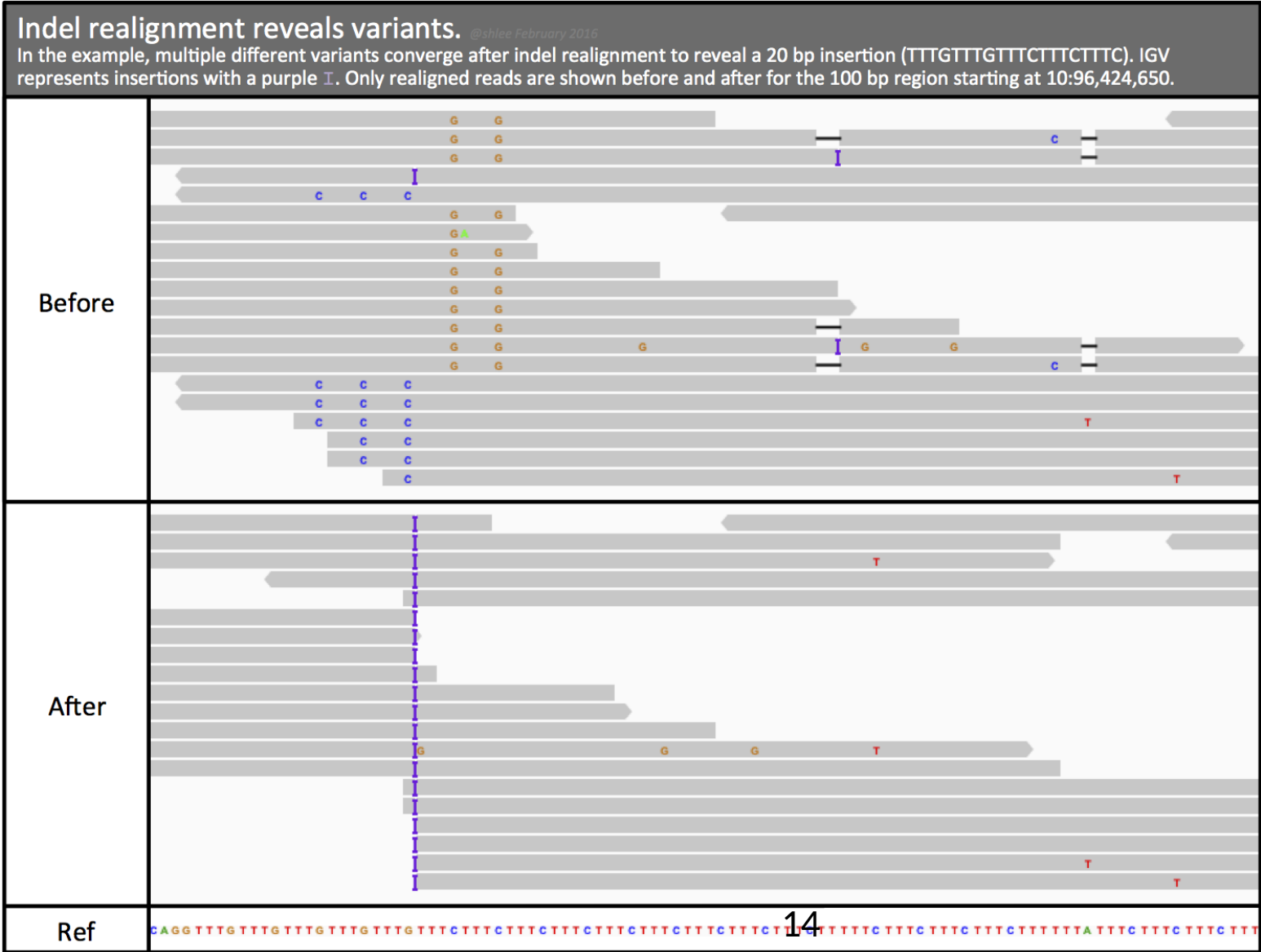
- Aligns sequencing reads to a reference genome.
- Balances speed and accuracy.
- Local realignment of all reads at a specific location simultaneously to minimize mismatches to the reference genome.
- Reduces erroneous SNPs refines location of INDELS

Variant Calling Workflow: Pre-processing

Pre-processing

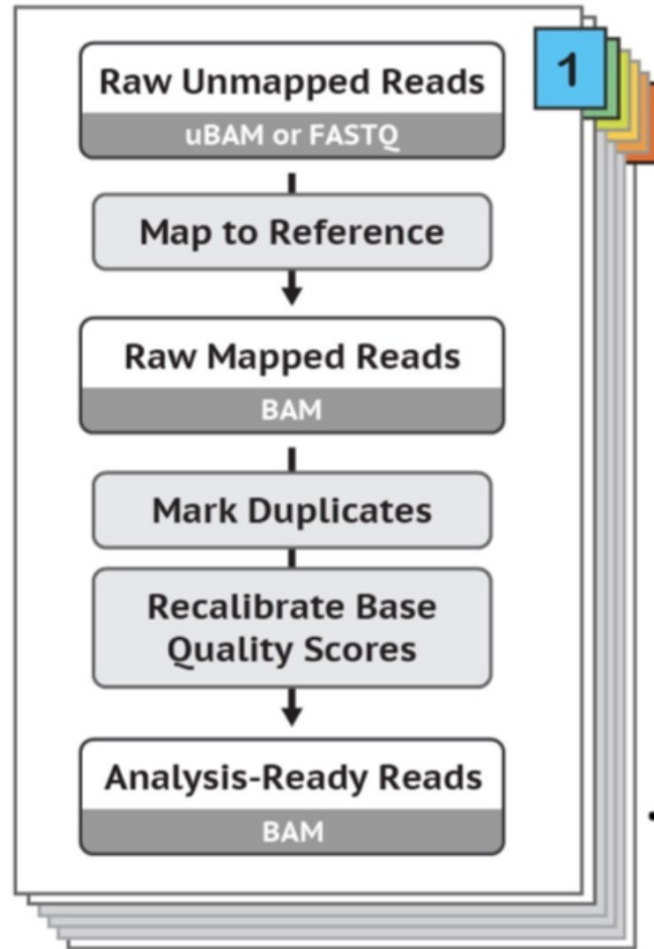


Local realignment



Variant Calling Workflow: Pre-processing

Pre-processing

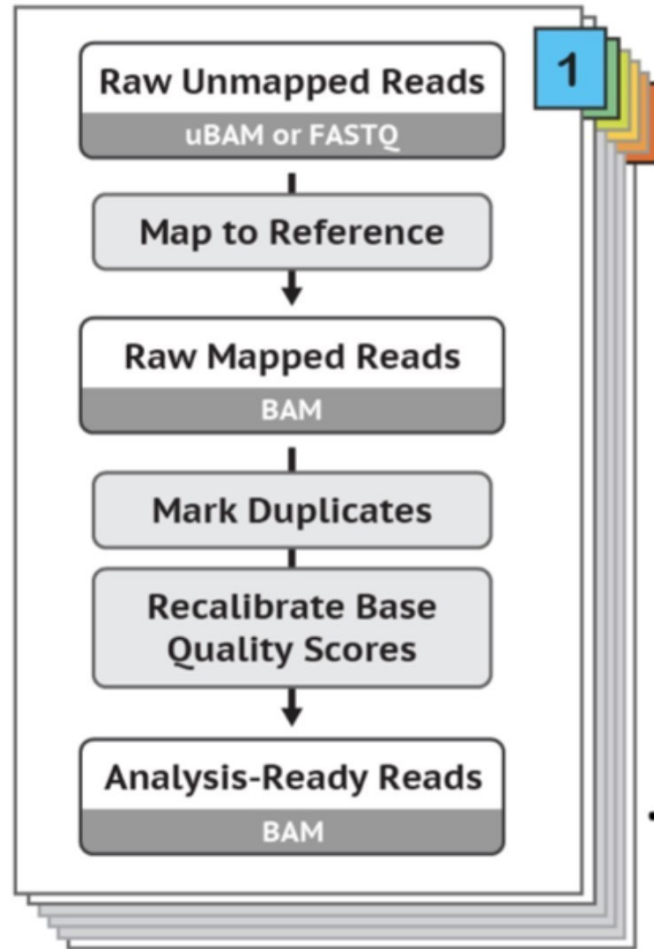


BWA-MEM - Read Alignment

- `bwa mem -M -t 4 -R "@RG\tID:SRR622461.7\tSM:NA12878\tLB:ERR194147\tPL:ILLUMINA"`
- `<reference> sample_1.fastq sample_2.fastq > alignment.sam`
- -M: inserts a tag to the alignment if non-primary alignment (required by GATK)
- -R: read group contains information such as the sample name, library and flow cell.
- -t: threads or number of cpus
- `<reference>`: path to reference genome in fasta format and the BWA index files

Variant Calling Workflow: Pre-processing

Pre-processing



BWA-MEM - Read Alignment

- Output is a SAM/BAM file.
- SAM file specifications: <https://samtools.github.io/hts-specs/SAMv1.pdf>

```
@RG      ID:SRR622461.7 SM:NA12878      LB:ERR194147      PL:ILLUMINA
```

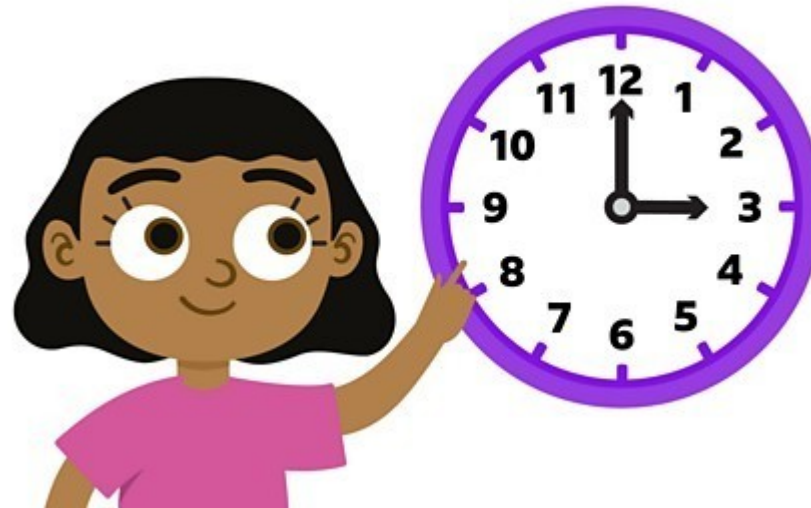
@RG → Indicates the start of a read group header.

ID:SRR622461.7 → Read group identifier (SRR622461.7 in this case), usually corresponding to a sequencing run or lane.

SM:NA12878 → Sample name (NA12878), which is a well-known human genome reference sample. LB:ERR194147 → Library name (ERR194147), referring to the sequencing library used for this sample. PL:ILLUMINA → Platform (ILLUMINA), specifying the sequencing

Alignment	position	CIGAR	read	flags/ metadata
ERR194147.45	163	chr18 6576006 99 101M = 6578028 317	CATTCT... <B<<BBBBB...	NM:i:0 MD:Z:101...
read name	flag	mapping quality	mate information ₁₆	PHRED quality

3 Minutes WhatsApp Break!!!



Variant Calling Workflow

1. Pre-processing

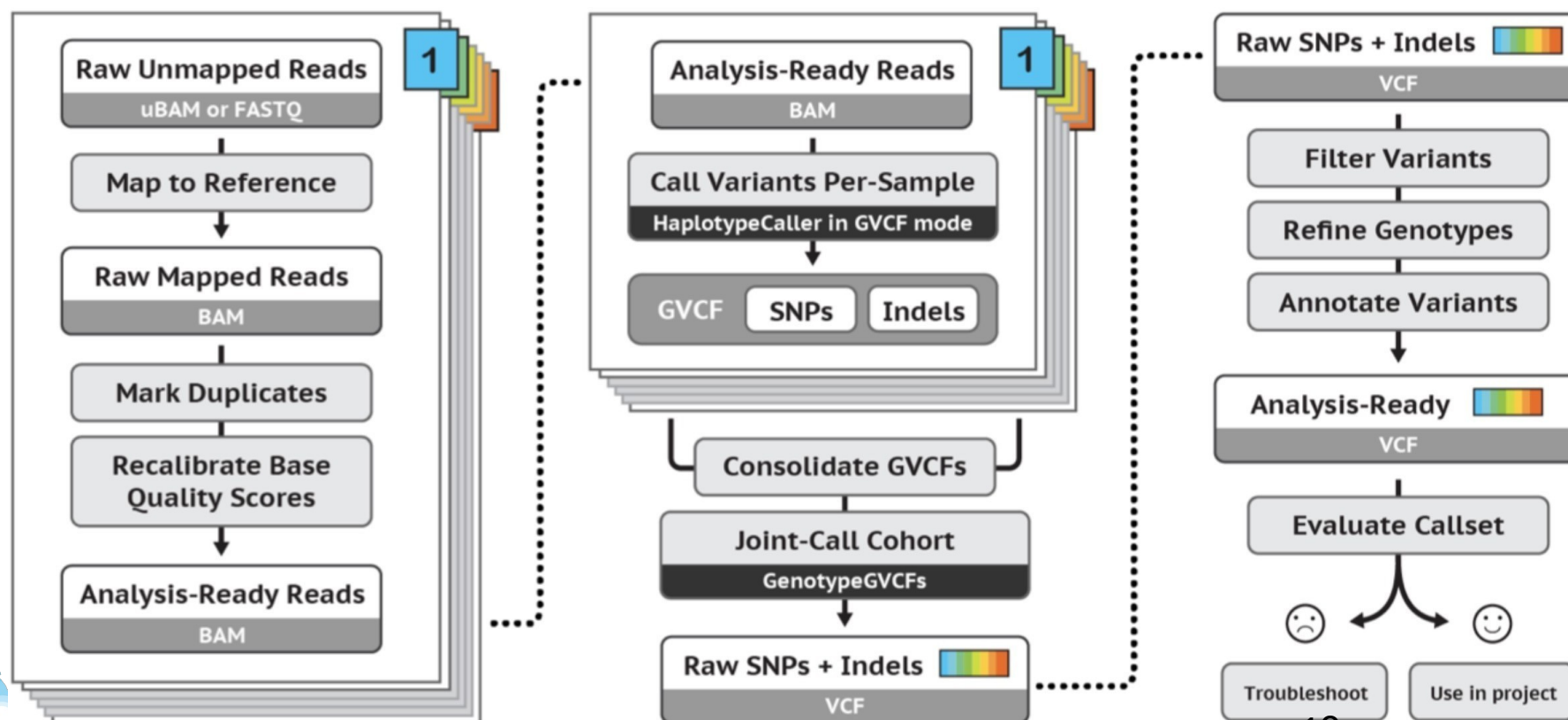
- Prepare analysis ready bam files

2. Variant discovery

- Variant discovery

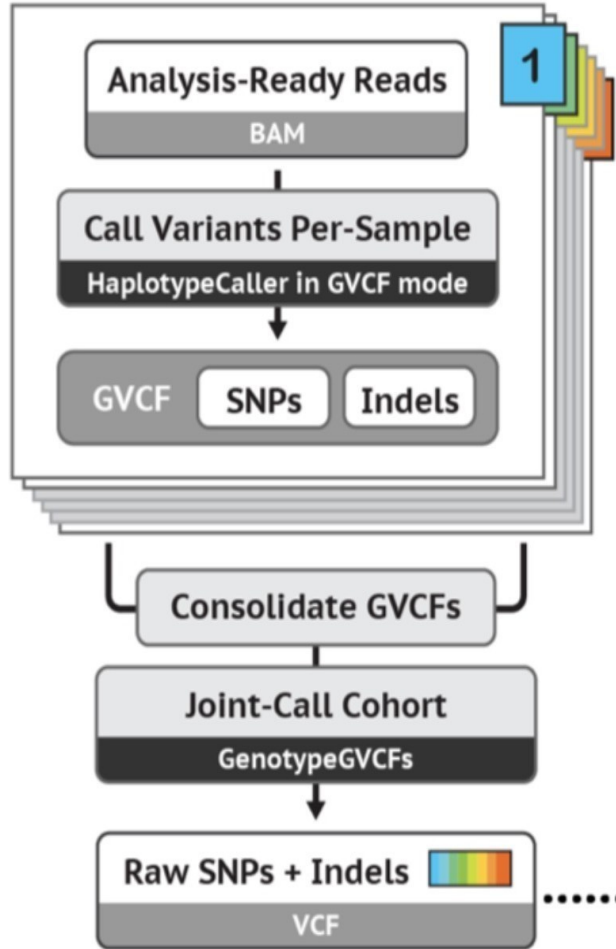
3. Refinement

- Use metadata from variant calling to improve genotype accuracy



Variant Calling Workflow: Variant Discovery

Variant Discovery

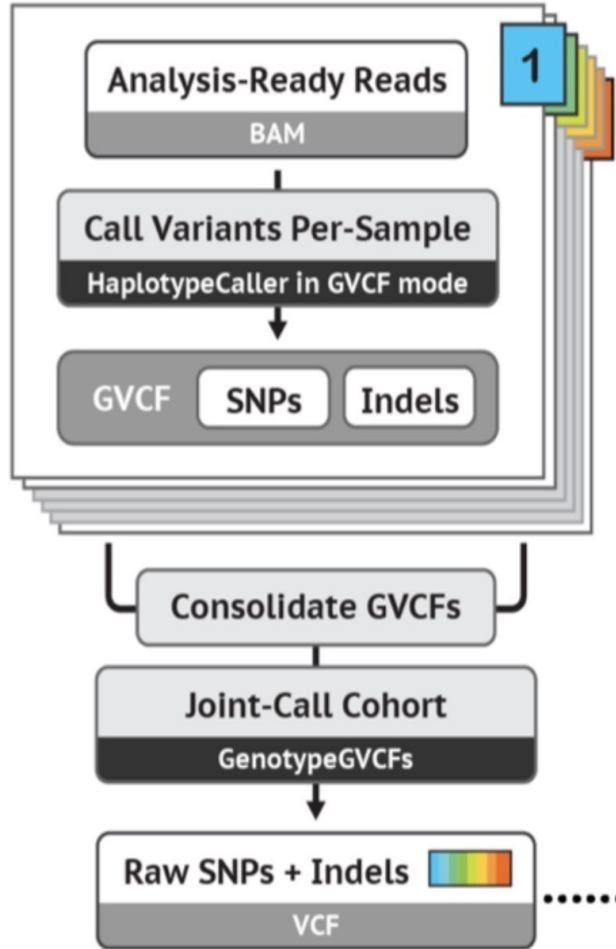


Variant Discovery

- Identifies Single Nucleotide Variants (SNVs) and INDELs.
- Uses probabilistic models and Bayesian approaches.
- Tools: GATK-HaplotypeCaller, Samtools/BCFtools, FreeBayes.

Variant Calling Workflow: Variant Discovery

Variant Discovery

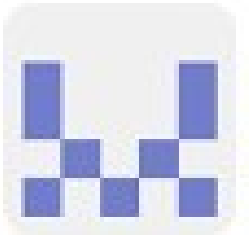


Samtools/BCFtools

- A lightweight tool for variant calling.
- Works well for small-scale variant calling.
- Provides filtering and annotation options.

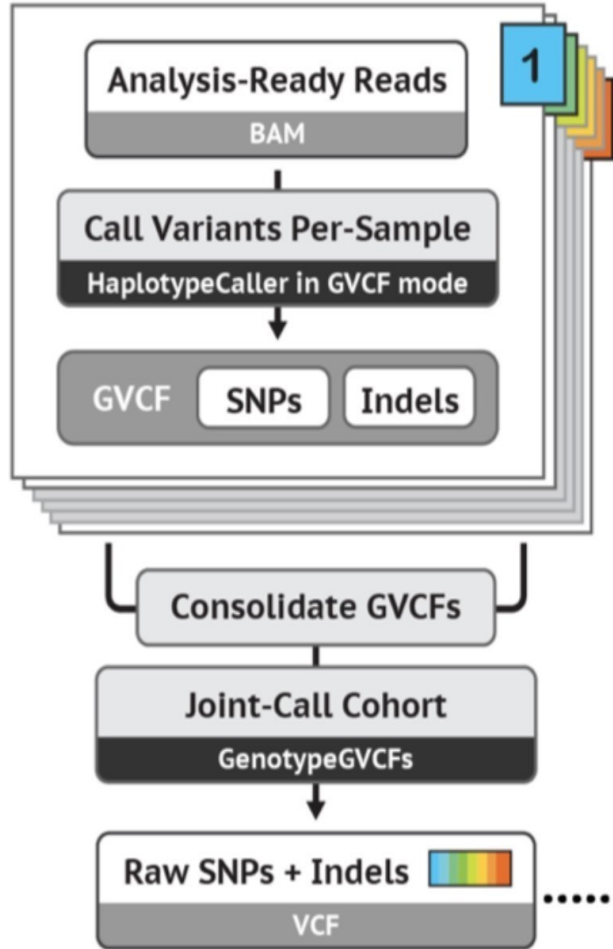
samtools/bcftools

This is the official development repository for BCFtools. See installation instructions and other documentation here
<http://samtools.github.io/bcftools/howtos/install.html>



Variant Calling Workflow: Variant Discovery

Variant Discovery



FreeBayes - Bayesian Variant Caller

- Uses a Bayesian approach for variant calling.
- Works well with pooled samples and polyploid genomes.
- Can call complex variants (e.g., multi-nucleotide polymorphisms).

freebayes/
freebayes

Bayesian haplotype-based genetic polymorphism discovery and genotyping.

AU 38
Contributors

41
Issues

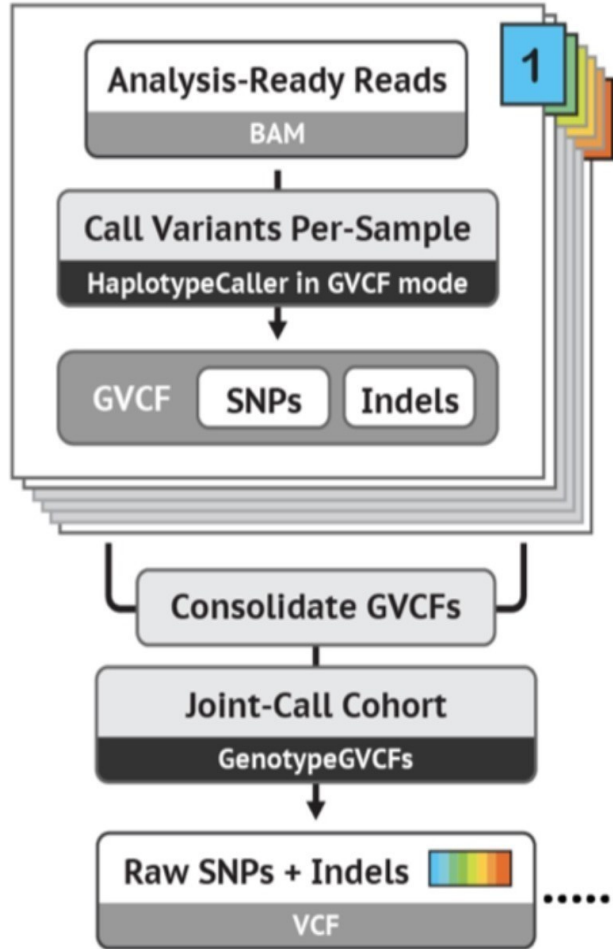
609
Stars

265
Forks

CATTCGGAT
CATTCGGAT
CATTCGGAT
CATTCGGAT
CATTCGGAT
CATTCGGAT

Variant Calling Workflow: Variant Discovery

Variant Discovery



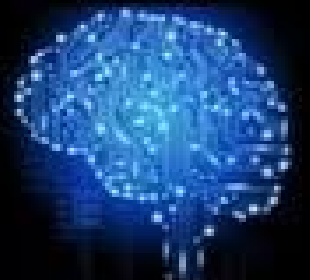
DeepVariant - AI-Based Variant Calling

- Uses deep learning for high-accuracy variant calling.
- Developed by Google.
- Outperforms traditional methods in difficult regions.

DeepVariant on DNAnexus

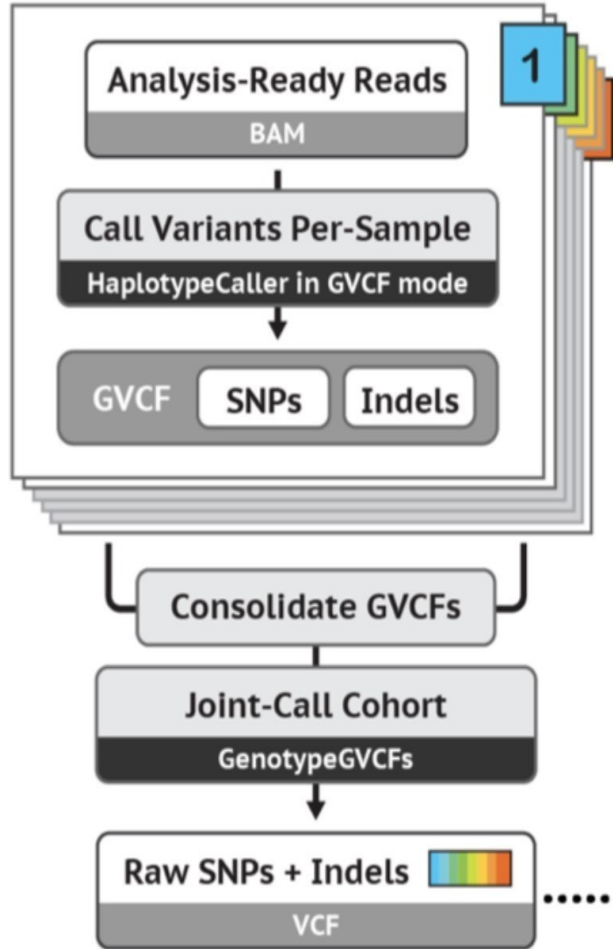
Google's Deep Learning-Based
Variant Caller

Request Access to Pilot Program



Variant Calling Workflow: Variant Discovery

Variant Discovery



Variant Discovery

- Output is a VCF file
- VCF file specifications <https://samtools.github.io/hts-specs/VCFv4.2.pdf>

Header

#fileformat=VCFv4.2

##FILTER=<ID=PASS,Description="All filters passed"> ##contig=<ID=1,length=249250621>

##FORMAT=<ID=AD,Number=R,Type=Integer,Description="Allelicdepths for the ref and alt alleles in the order listed">

Variant record

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	Sample1
1	567376	.	G	A	146.3	PASS	AC=1;DP=55	GT:AD:DP	0/1:30,25:55

Variant Calling Workflow

1. Pre-processing

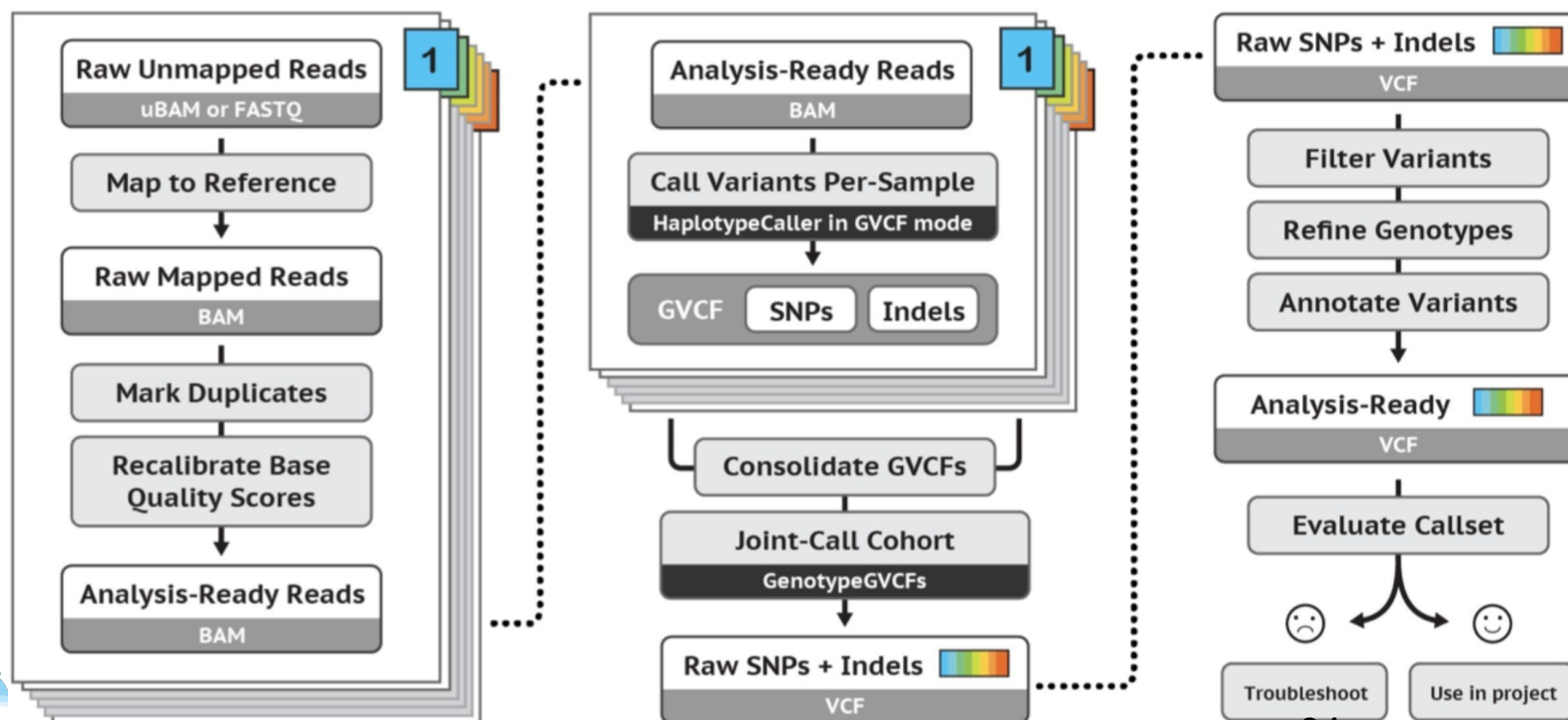
- Prepare analysis ready bam files

2. Variant discovery

- Variant discovery

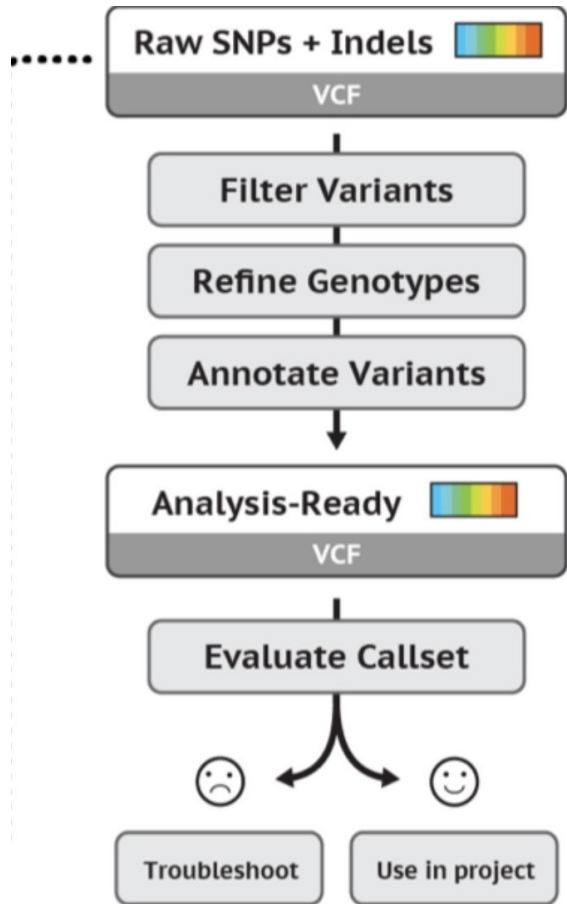
3. Refinement

- Use metadata from variant calling to improve genotype accuracy



Variant Calling Workflow: Refinement

Refinement

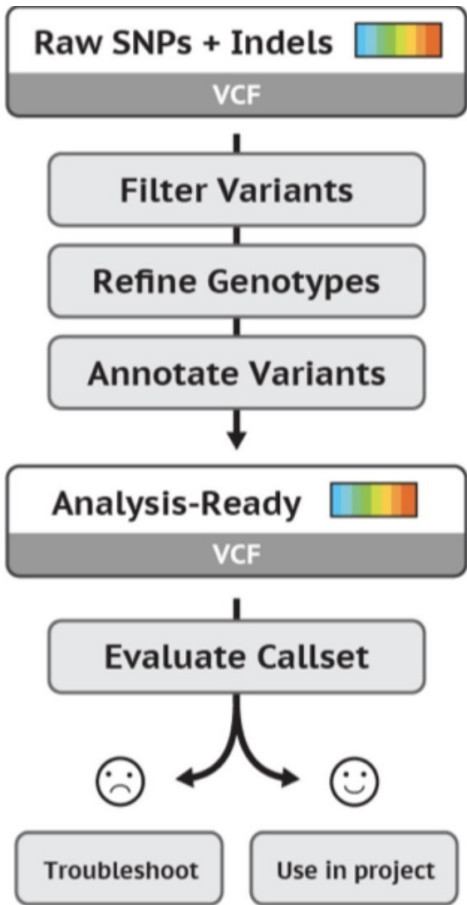


- Variant callers are sensitive
- Removes low-quality and false-positive variants.
- Strategies include:
 - Variant quality score recalibration (using known sites)
 - Hard filtering on quality criteria
 - Annotation features
- Ensures high-confidence variant calls.
- Tools: VCFtools, GATK VariantFiltration, SnpEff

Variant Calling Workflow:

Refinement

Refinement



VCF format

##[HEADER LINES]										
#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	NA12878	
1	873762	.	T	G	5231.78	PASS	[ANNOTATIONS]	GT:AD:DP:GQ:PL	0/1:173,141:282:99:255,0,255	
1	877664	rs3828047	A	G	3931.66	PASS	[ANNOTATIONS]	GT:AD:DP:GQ:PL	1/1:0,105:94:99:255,255,0	
1	899282	rs28548431	C	T	71.77	PASS	[ANNOTATIONS]	GT:AD:DP:GQ:PL	0/1:1,3:4:25.92:103,0,26	
1	974165	rs9442391	T	C	29.84	LowQual	[ANNOTATIONS]	GT:AD:DP:GQ:PL	0/1:14,4:14:60.91:61,0,255	

INFO fields:

GT: Genotype - Indicates the alleles at the variant site for each sample separated by (/). 0 for the reference allele, 1 for the first alternate allele

AD: Allele Depth - The count of reads supporting each allele. if AD is 1,3, it means there are 1 reads supporting the reference allele and 3 reads supporting the alternate allele.

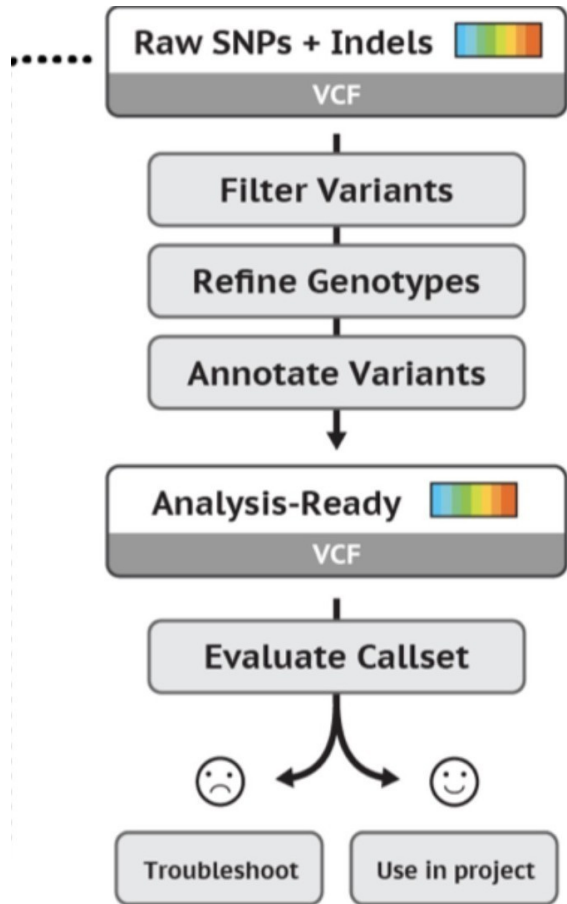
DP: Depth of Coverage - The total count of reads that cover this genomic position across all alleles

GQ: Genotype Quality - a Phred-scaled quality score for the assertion made in GT

PL: Phred-scaled Likelihoods of Genotypes - Represents the likelihood of each genotype given the sequencing data, converted to a Phred scale (i.e., $-10 * \log_{10}(\text{likelihood})$).

Variant Calling Workflow: Refinement

Refinement



False positives and negatives

• FALSE POSITIVES:

- Error rate: 1/10,000 bases
- False positives can be decreased by increasing the number of control and cases samples.

• FALSE NEGATIVES:

- Variants not sequenced → missing values and errors in variants
- Two types of missing values:
 1. No coverage: $./.$  Not sequenced base
 2. Filtered: $-/-$  Low quality base

Variant Calling Workflow: Refinement

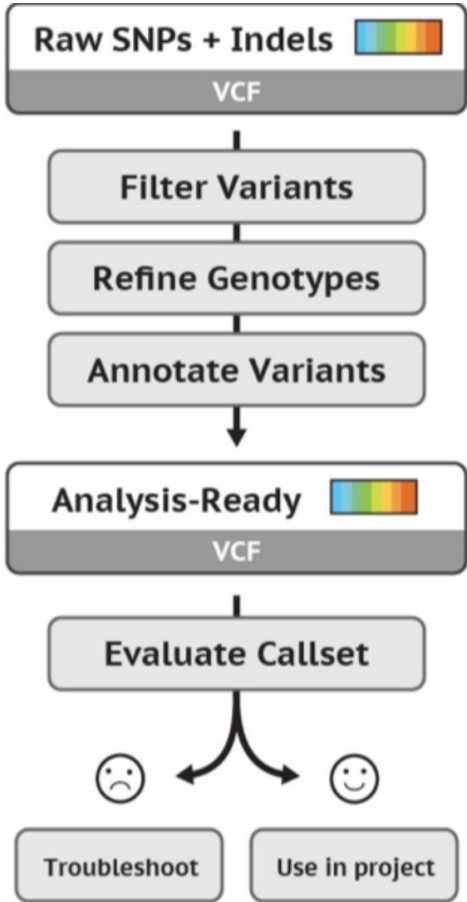
Refinement

Filtering SNVs and indels

- Filtering parameters for SNVs and indels are different.

SNVs
QD < 2.0
MQ < 40.0
FS > 60.0
ReadPosRankSum < -8.0

INDELS
QD < 2.0
FS > 200.0
ReadPosRankSum < -20.0



##[HEADER LINES]

#CHR OM	POS	ID	RE F	AL T	QUAL	FILTE R	INFO	FORMAT	NA12878	
1	87376 2	.	T	G	521.7 8	PASS	[ANNOTATION S]	GT:AD:DP:GQ: PL	0/1:173,141:282:99:255,0,255	
1	87766 4	rs382804	7	A	G	3931.6 6	PASS	[ANNOTATION S]	GT:AD:DP:GQ: PL	1/1:0,105:94:99:255,255,0
1	89928 2	rs285484		C	T	71.77	PASS	[ANNOTATION S]	GT:AD:DP:GQ: PL	0/1:1,3:4:25.92:103,0,26
1	97416 5	rs944239	1	T	C	29.84	LowQual	[ANNOTATION S]	GT:AD:DP:GQ: PL	0/1:14,4:14:60.91:61,0,255

Variant Calling Workflow: Refinement

Filtering SNVs and indels

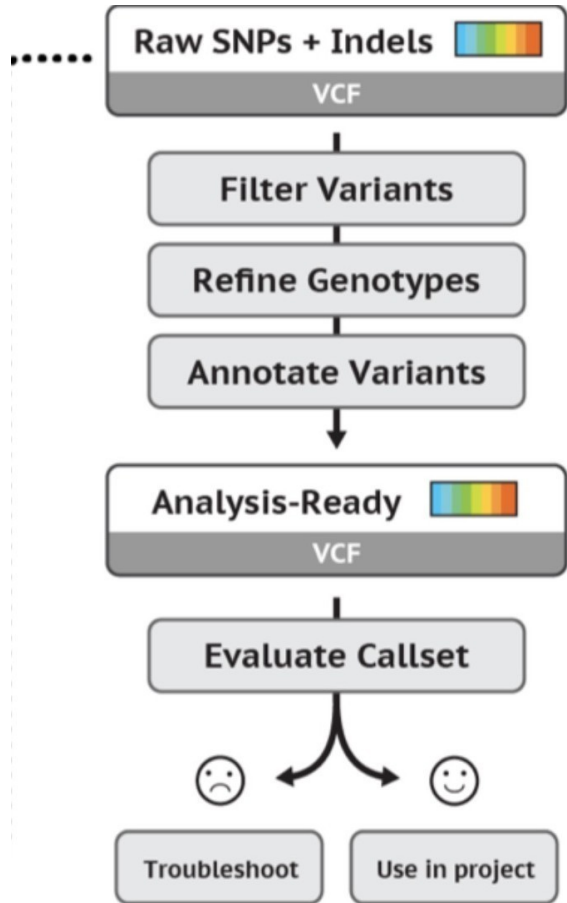
- Filtering parameters for SNVs and indels are different.

Filter	Stands For	What It Indicates
QD (Quality by Depth)	Variant confidence / dep	Low QD (<2.0) = uncertain call relative to depth often a weak variant signal
MQ (Mapping Quality)	RMS of mapping qualities	Low MQ (<40.0) = reads supporting the variant are poorly aligned
FS (Fisher Strand Bias)	Strand bias	High FS (>60.0 or >200.0) = variant only appears on one strand (possibly an artifact)
ReadPosRankSum	Read position bias	Negative values = variant is often near read ends, which is error-prone

Variant Calling Workflow: Refinement

Refinement

Annotation of Variants



- Provides biological context for variants.
- Identifies functional effects and disease associations.

Strategies of Annotation of the variants:

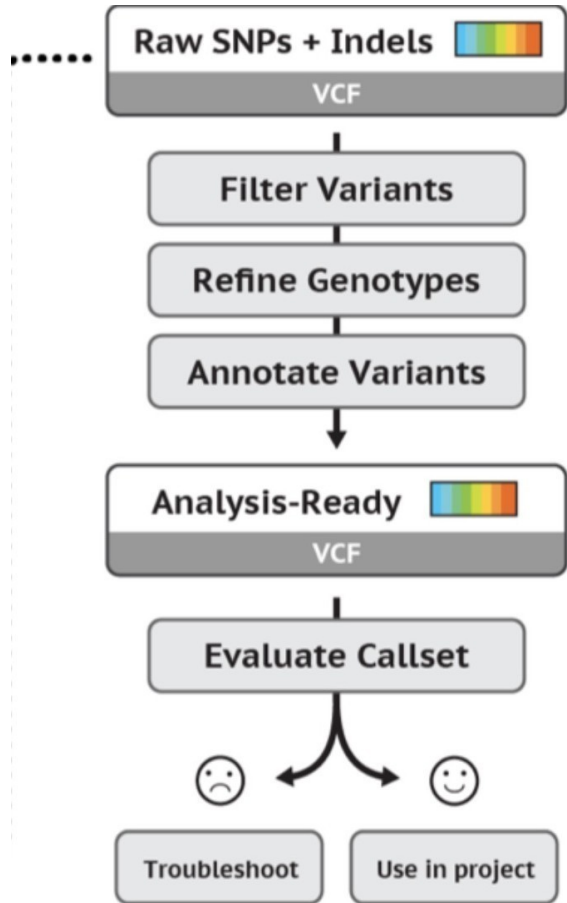
- gene → RefSeq
- consequence type.
- PolyPhen.
- SIFT.

Tools: SnpEff, Annovar, VEP.

Variant Calling Workflow: Refinement

Refinement

Annotation of Variants

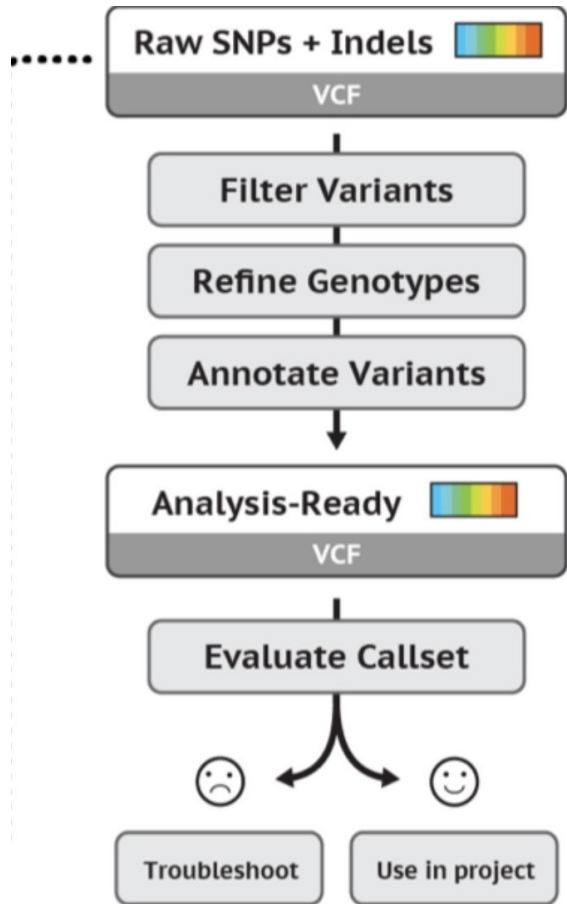


- **SnEff - Variant Annotation**

- It can annotate single nucleotide variants (SNVs) and insertions/deletions.
- Very fast.
- Predicts functional effects of genetic variants.
- Annotates with known variant databases.
- Useful for prioritizing clinically relevant variants.

Variant Calling Workflow

Refinement



Fully Built Pipelines - Snippy

- Rapid tool designed for calling variants from next-generation sequencing data.
- Used in microbial genomics, particularly for bacteria, due to its efficiency and effectiveness.

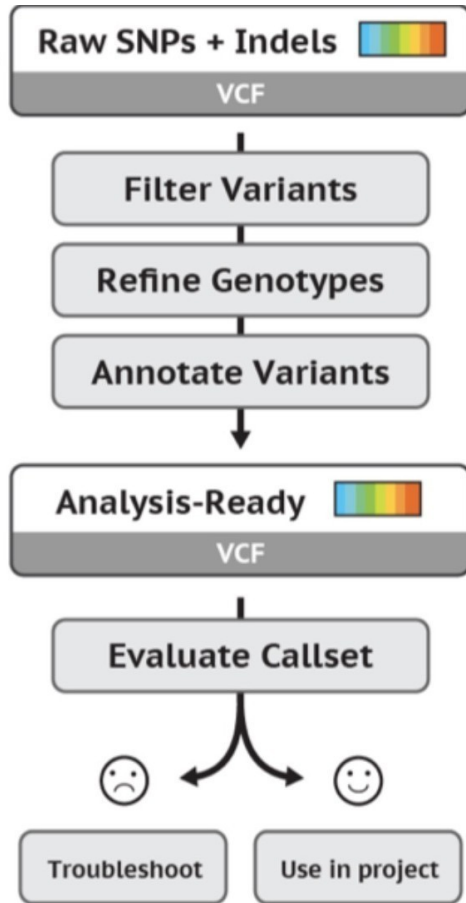
Individual Tools contained:

- **Bwa-mem** - Aligns reads to the reference genome
- **Samtools** - Sorts and indexes BAM files
- **Freebayes** - Calls variants (SNPs, indels) from alignments

SnEff – Annotation of variants

Variant Calling Workflow: Refinement

Refinement



Visualization Tools

- Helps interpret sequencing data and variants.
- Tools: Integrative Genomics Viewer (IGV), UCSC Genome Browser, Jalview, etc.



Variant Calling Workflow

1. Pre-processing

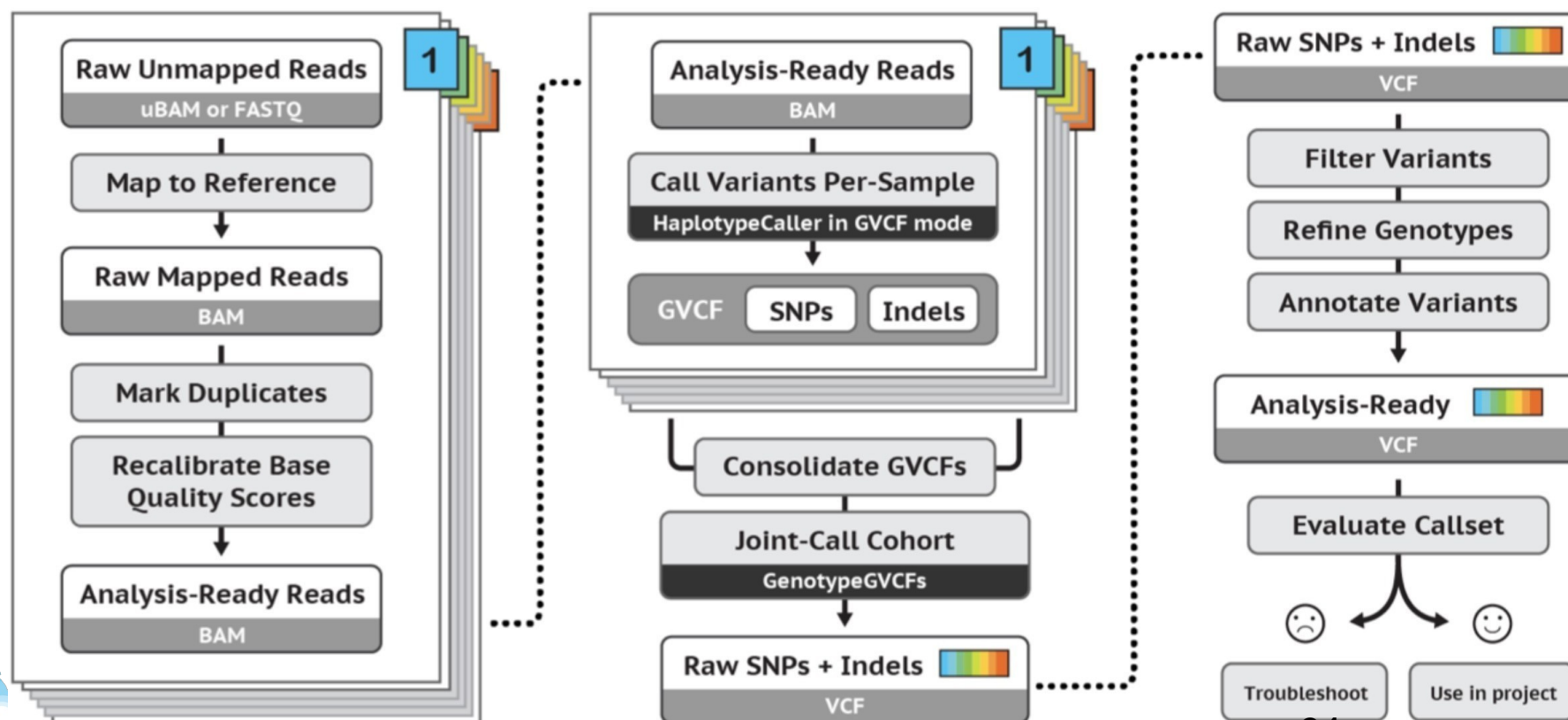
- Prepare analysis ready bam files

2. Variant discovery

- Variant discovery

3. Refinement

- Use metadata from variant calling to improve genotype accuracy



Hands On

From the E. coli trimmed raw fastq data on, perform variant calling and annotation.

Use the **K-12 MG** strain as the reference.